

Data Science & Exploratory Data Analysis

A PRESENTATION OF INTERNSHIP SUBMITTED BY :

TOPIC: SALES INSIGHTS & CANCER PREDICTION

NAME : VRUSHABH BODARYA

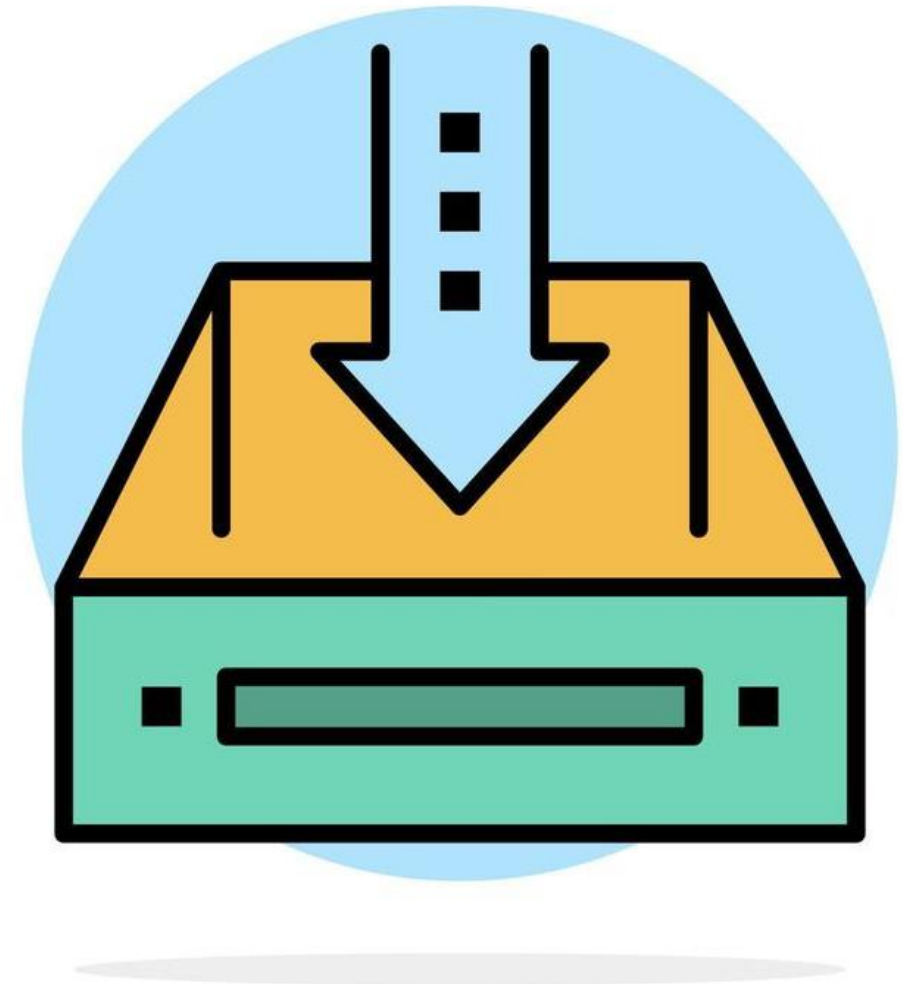
ENROLLMENT NO. : 200020116035

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**AT THE COMPANY : APPSTONE LAB TECHNOLOGIES
LLP**

ABSTRACT

- This project integrates data analysis for sales insights and breast cancer prediction using data science techniques within a unified framework built on Streamlit. Leveraging comprehensive data analysis methodologies, including exploratory data analysis and machine learning algorithms, we aim to extract actionable insights from sales data and develop accurate forecasting models to anticipate future sales trends.



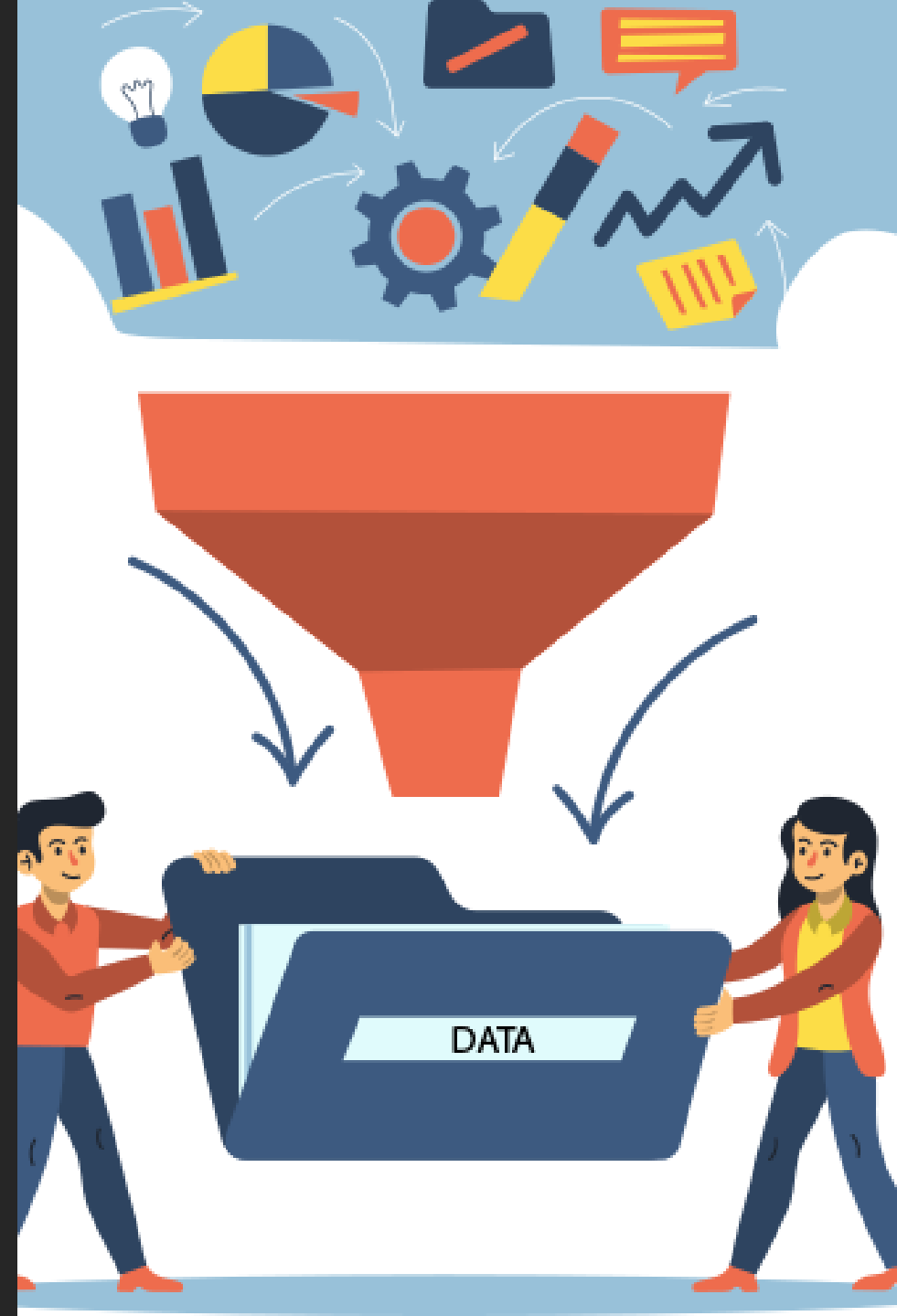
Disclose Sales Insights with Data Analysis, Power BI and Time Series Forecasting

- Welcome to our presentation on finding sales insights using data analysis, Power BI, and time series analysis.
- In today's competitive market, understanding sales trends and forecasting future sales is essential for business success.
- This presentation explores how we can leverage data analysis techniques, Power BI and time series analysis to gain valuable insights into sales data.



Data Collection & Preparation

- ❖ The first step in finding sales insights is collecting and preparing the data.
- ❖ Data sources may include sales transactions, customer demographics, marketing campaigns, and external factors like economic indicators.
- ❖ Cleaning, transforming, and integrating the data ensure its quality and consistency for analysis.



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Conditional Formatting

Format as Table

Cell Styles

Styles

Insert

Delete

Format

Cells

AutoSum

Fill

Clear

Editing

Sort & Filter

Find & Select

Add-ins

A1

Row ID+O6G3A1:R6

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Row ID+O	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer	Segment	Country	City	State	Region	Product ID	Category	Sub-Categ	Product Na	Sales	Quantity	Profit	Returns	Payment M	ind1	ind2
2	4918	CA-2019-1	#####	#####	Standard C	BM-11575	Brendan M	Corporate	United Sta	Gaithersbu	Maryland	East	FUR-BO-10	Furniture	Bookcases	Bush West	73.94	1	28.2668	#N/A	Online		
3	4919	CA-2019-1	#####	#####	Standard C	BM-11575	Brendan M	Corporate	United Sta	Gaithersbu	Maryland	East	FUR-BO-10	Furniture	Bookcases	Bush West	173.94	3	38.2668	#N/A	Online		
4	4920	CA-2019-1	#####	#####	Standard C	BM-11575	Brendan M	Corporate	United Sta	Gaithersbu	Maryland	East	TEC-PH-10	Technolog	Phones	GE 30522E	231.98	2	67.2742	#N/A	Cards		
5	3074	CA-2019-1	#####	#####	First Class	LR-16915	Lena Radf	Consumer	United Sta	Los Angele	California	West	OFF-ST-10	Office Sup	Storage	Recycled S	114.46	2	28.615	#N/A	Online		
6	8604	US-2019-1	#####	#####	Standard C	CA-12310	Christine A	Corporate	United Sta	San Antoni	Texas	Central	TEC-AC-10	Technolog	Accessorie	Imation Cl	30.08	2	-5.264	#N/A	Online		
7	8605	US-2019-1	#####	#####	Standard C	CA-12310	Christine A	Corporate	United Sta	San Antoni	Texas	Central	TEC-AC-10	Technolog	Accessorie	WD My Pa	165.6	3	-6.21	#N/A	Online		
8	8606	US-2019-1	#####	#####	Standard C	CA-12310	Christine A	Corporate	United Sta	San Antoni	Texas	Central	TEC-PH-10	Technolog	Phones	AT&T 1792	180.96	5	13.572	#N/A	Cards		
9	9494	CA-2019-1	#####	#####	Standard C	BO-11350	Bill Overfe	Corporate	United Sta	Broken Arr	Oklahoma	Central	FUR-TA-10	Furniture	Tables	Hon Practi	1592.85	7	350.427	#N/A	COD		
10	9495	CA-2019-1	#####	#####	Standard C	BO-11350	Bill Overfe	Corporate	United Sta	Broken Arr	Oklahoma	Central	OFF-BI-10	Office Sup	Binders	Storex Dur	11.88	2	5.346	#N/A	COD		
11	2898	US-2019-1	#####	#####	Standard C	EB-13975	Erica Bern	Corporate	United Sta	Charlotte	North Car	South	TEC-CO-10	Technolog	Copiers	Hewlett Pa	959.968	4	119.996	#N/A	Online		
12	5868	CA-2019-1	#####	#####	Standard C	BP-11185	Ben Peterr	Corporate	United Sta	Philadelph	Pennsylv	East	OFF-AR-10	Office Sup	Art	Newell 31	4.672	1	0.584	#N/A	Online		
13	5869	CA-2019-1	#####	#####	Standard C	BP-11185	Ben Peterr	Corporate	United Sta	Philadelph	Pennsylv	East	OFF-BI-10	Office Sup	Binders	Avery Arch	104.58	6	-80.178	#N/A	Online		
14	863	CA-2019-1	#####	#####	Second Cl	AJ-10795	Anthony Jc	Corporate	United Sta	Jacksonvill	Florida	South	TEC-AC-10	Technolog	Accessorie	Logitech M	191.472	6	40.6878	#N/A	COD		
15	864	CA-2019-1	#####	#####	Second Cl	AJ-10795	Anthony Jc	Corporate	United Sta	Jacksonvill	Florida	South	OFF-AR-10	Office Sup	Art	Newell 33	5.248	2	0.5904	#N/A	Online		
16	865	CA-2019-1	#####	#####	Second Cl	AJ-10795	Anthony Jc	Corporate	United Sta	Jacksonvill	Florida	South	TEC-PH-10	Technolog	Phones	Logitech B	59.184	2	5.1786	#N/A	COD		
17	2162	CA-2019-1	#####	#####	Standard C	DW-13480	Dianna Wi	Home Offi	United Sta	Oakland	California	West	OFF-AR-10	Office Sup	Art	Panasonic	34.58	1	10.0282	#N/A	COD		
18	8031	CA-2019-1	#####	#####	Standard C	NM-18520	Neoma M	Consumer	United Sta	Amarillo	Texas	Central	FUR-FU-10	Furniture	Furnishings	Executive	23.076	3	-10.9611	#N/A	COD		
19	8032	CA-2019-1	#####	#####	Standard C	NM-18520	Neoma M	Consumer	United Sta	Amarillo	Texas	Central	OFF-PA-10	Office Sup	Paper	Xerox 212	25.92	5	9.072	#N/A	Cards		
20	6851	US-2019-1	#####	#####	Standard C	JO-15145	Jack O'Bria	Corporate	United Sta	Franklin	Wisconsin	Central	FUR-BO-10	Furniture	Bookcases	Atlantic M	1565.88	6	407.1288	#N/A	Online		
21	6852	US-2019-1	#####	#####	Standard C	JO-15145	Jack O'Bria	Corporate	United Sta	Franklin	Wisconsin	Central	OFF-BI-10	Office Sup	Binders	Plastic Bin	106.05	7	49.8435	#N/A	Cards		
22	7808	US-2019-1	#####	#####	Standard C	VS-21820	Vivek Sund	Consumer	United Sta	Raleigh	North Car	South	OFF-BI-10	Office Sup	Binders	XtraLife Cl	30.828	7	-24.6624	#N/A	Cards		
23	7809	US-2019-1	#####	#####	Standard C	VS-21820	Vivek Sund	Consumer	United Sta	Raleigh	North Car	South	OFF-AR-10	Office Sup	Art	Newell 31	47.616	3	5.952	#N/A	COD		
24	7810	US-2019-1	#####	#####	Standard C	VS-21820	Vivek Sund	Consumer	United Sta	Raleigh	North Car	South	TEC-PH-10	Technolog	Phones	Avaya 541	108.784	2	10.8784	#N/A	COD		
25	3209	CA-2019-1	#####	#####	Standard C	LA-16780	Laura Arm	Corporate	United Sta	Fresno	California	West	TEC-AC-10	Technolog	Accessorie	Logitech G	349.95	5	118.983	#N/A	COD		
26	3210	CA-2019-1	#####	#####	Standard C	LA-16780	Laura Arm	Corporate	United Sta	Fresno	California	West	TEC-PH-10	Technolog	Phones	netTALK D	377.928	9	141.723	#N/A	Cards		
27	3702	CA-2019-1	#####	#####	Second Cl	MF-17665	Maureen F	Corporate	United Sta	Toledo	Ohio	East	FUR-FU-10	Furniture	Furnishings	DAX Two-1	15.168	2	3.792	#N/A	Cards		
28	4667	CA-2019-1	#####	#####	Standard C	KR-16585	Kay Black	Corporate	United Sta	Seattle	Washington	West	FUR-FU-10	Furniture	Furnishings	24 Hour P	70.92	4	24.2556	#N/A	Cards		

```
In [22]: # Insert all require libraries
```

```
In [1]: import pandas as pd
```

```
In [3]: import matplotlib.pyplot as plt
```

```
In [4]: import seaborn as sns
```

```
In [6]: # read csv file
walmart = pd.read_csv('SuperStore_Sales_Dataset.csv')
```

```
In [7]: # provide top 5 rows of data
walmart.head()
```

```
Out[7]:
```

	Row ID+O6G3A1:R6	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Category	Sub- Category	Product Name	Sales	Quant
0	4918	CA- 2019- 160304	01- 01- 2019	07- 01- 2019	Standard Class	BM-11575	Brendan Murry	Corporate	United States	Gaithersburg	...	Furniture	Bookcases	Bush Westfield Collection Bookcases, Medium Ch...	73.94	
1	4919	CA- 2019- 160304	02- 01- 2019	07- 01- 2019	Standard Class	BM-11575	Brendan Murry	Corporate	United States	Gaithersburg	...	Furniture	Bookcases	Bush Westfield Collection Bookcases, Medium Ch...	173.94	
2	4920	CA- 2019- 160304	02- 01- 2019	07- 01- 2019	Standard Class	BM-11575	Brendan Murry	Corporate	United States	Gaithersburg	...	Technology	Phones	GE 30522EE2	231.98	
3	3074	CA- 2019- 125206	03- 01- 2019	05- 01- 2019	First Class	LR-16915	Lena Radford	Consumer	United States	Los Angeles	...	Office Supplies	Storage	Recycled Steel Personal File for Hanging File ...	114.46	
4	8604	US- 2019- 116365	03- 01- 2019	08- 01- 2019	Standard Class	CA-12310	Christine Abelman	Corporate	United States	San Antonio	...	Technology	Accessories	Imation Clip USB flash drive - 8 GB	30.08	

Projects/EDA on Walmart Sales Forecasting

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EDA on Walmart Sales Forecasting

Last Checkpoint: 03/05/2024 (autosaved)

Logout

FileEditViewInsertCellKernelWidgetsHelpNot TrustedPython 3 (ipykernel)

Run

Code

Data Pre-processing

In [20]:

removing ind1, ind2 column
df = walmart.drop(['ind1','ind2'],axis=1)

In [24]:

df.head()

Out[24]:

	Row ID+O6G3A1:R6	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Region	Product ID	Category	Sub-Category	Pro N
0	4918	CA-2019-160304	01-01-2019	07-01-2019	Standard Class	BM-11575	Brendan Murry	Corporate	United States	Gaithersburg	...	East	FUR-BO-10004709	Furniture	Bookcases	Wes' Colle Bookca Me
1	4919	CA-2019-160304	02-01-2019	07-01-2019	Standard Class	BM-11575	Brendan Murry	Corporate	United States	Gaithersburg	...	East	FUR-BO-10004709	Furniture	Bookcases	Wes' Colle Bookca Me
2	4920	CA-2019-160304	02-01-2019	07-01-2019	Standard Class	BM-11575	Brendan Murry	Corporate	United States	Gaithersburg	...	East	TEC-PH-10000455	Technology	Phones	30522
3	3074	CA-2019-125206	03-01-2019	05-01-2019	First Class	LR-16915	Lena Radford	Consumer	United States	Los Angeles	...	West	OFF-ST-10003692	Office Supplies	Storage	Recy S Pers Fil Har F
4	8604	US-2019-116365	03-01-2019	08-01-2019	Standard Class	CA-12310	Christine Abelman	Corporate	United States	San Antonio	...	Central	TEC-AC-10002217	Technology	Accessories	Imation USB driv

34°C Mostly sunny

Search

14:30 19-03-2024

Data Cleaning

Checking For Duplicates

```
In [31]: # Using Conditional Statement

if df.duplicated().sum()>0:
    print('duplicates are present')
else:
    print('duplicates are not exist')
```

duplicates are not exist

```
In [32]: df.duplicated()
```

```
Out[32]: 0      False
1      False
2      False
3      False
4      False
...
5896   False
5897   False
5898   False
5899   False
5900   False
Length: 5901, dtype: bool
```

```
In [33]: df.duplicated(keep=False).sum()
```

```
Out[33]: 0
```



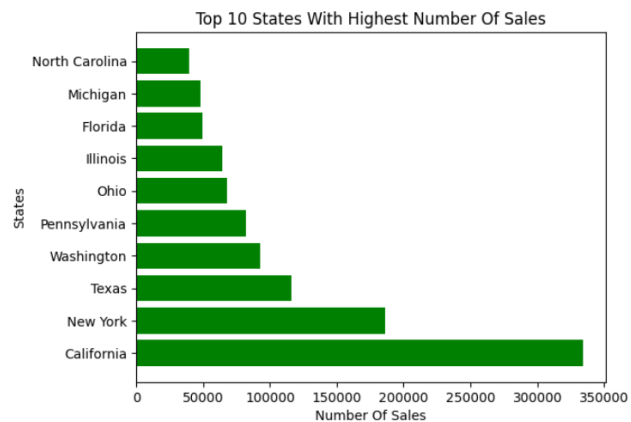
Exploratory Data Analysis And Power BI

- ❖ Power BI is a powerful business analytics tool for visualizing and exploring data.
- ❖ EDA techniques such as data visualization, filtering, and drill-down analysis help uncover patterns and trends in sales data.
- ❖ Interactive dashboards and reports provide stakeholders with actionable insights in real-time.

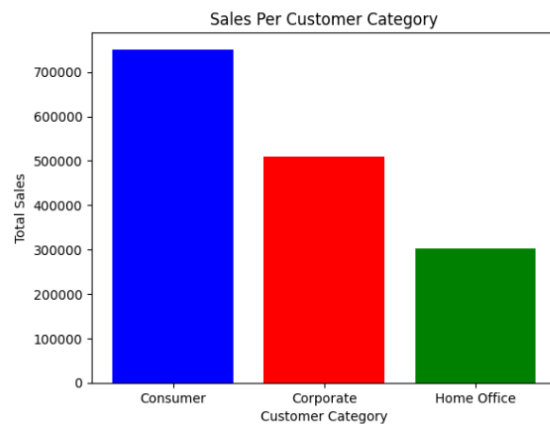


In [111]: # create Bar graph

```
plt.barh(Top_State_Sales['State'],Top_State_Sales['Sales'],color='green')
plt.title("Top 10 States With Highest Number Of Sales")
plt.xlabel("Number Of Sales")
plt.ylabel("States")
plt.show()
```



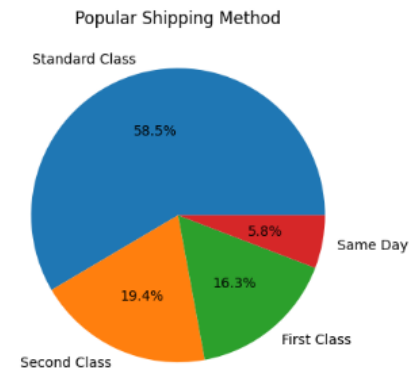
In [48]: plt.bar(sales_per_category['Segment'],sales_per_category['Sales'],color=['blue','red','green'])
plt.title('Sales Per Customer Category')
plt.xlabel('Customer Category')
plt.ylabel('Total Sales')
plt.show()



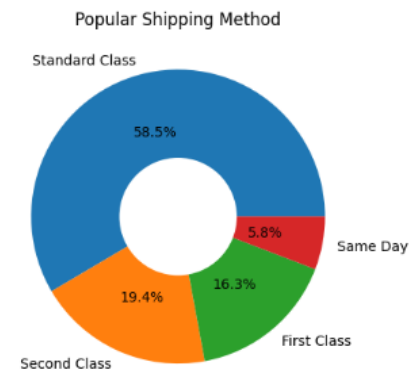
In [55]: df['Ship Mode'].value_counts(normalize=True)

```
Out[55]: Ship Mode
Standard Class    0.584816
Second Class     0.194374
First Class       0.162515
Same Day          0.058295
Name: proportion, dtype: float64
```

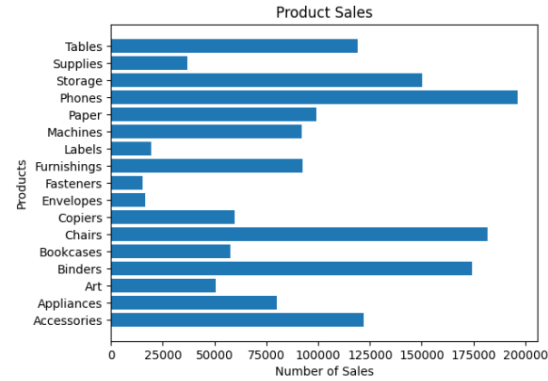
In [59]: # create a pie chart
plt.pie(Shipping_mode['count'],labels=Shipping_mode['Ship Mode'],autopct='%1.1f%%')
plt.title('Popular Shipping Method')
plt.show()



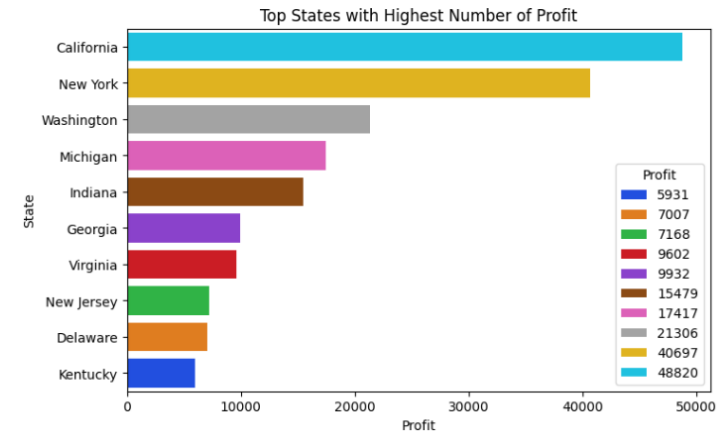
In [63]: # create a donut-pie chart
plt.pie(Shipping_mode['count'],labels=Shipping_mode['Ship Mode'],autopct='%1.1f%%',radius=1)
plt.pie([1],radius=0.4,colors=['w'])
plt.title("Popular Shipping Method")
plt.show()



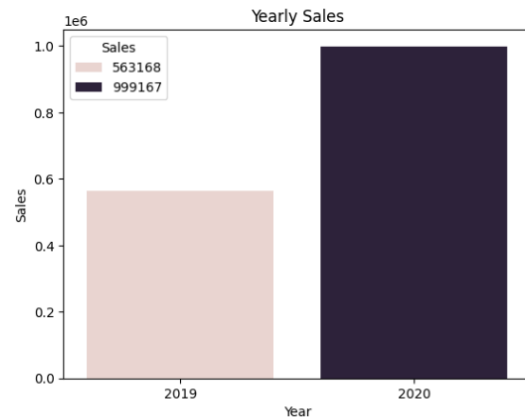
```
In [140]: # plotting bargraph
plt.barh(product_sales['Product'],product_sales['Sales'])
plt.title('Product Sales')
plt.ylabel('Products')
plt.xlabel('Number of Sales')
plt.show()
```



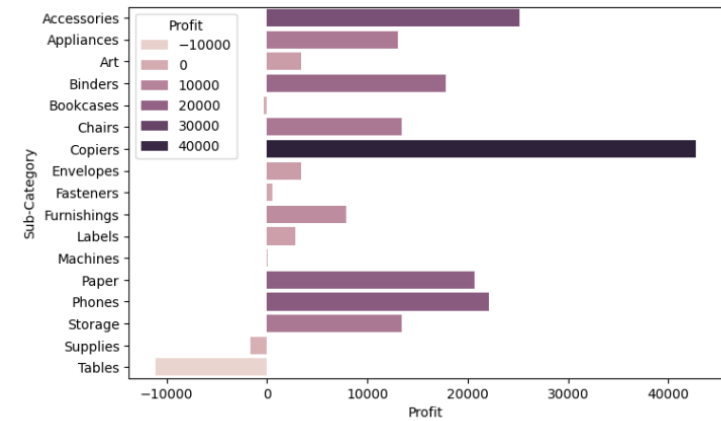
```
In [187]: plt.figure(figsize=(8,5))
sns.barplot(x='Profit',y='State',data=top_state,palette='bright',hue='Profit')
plt.title("Top States with Highest Number of Profit")
plt.show()
```



```
In [157]: # plotting bar graph based on yearly sales
sns.barplot(x='Year',y='Sales',data=yearly_sales,hue='Sales')
plt.title('Yearly Sales')
plt.show()
```



```
In [201]: plt.figure(figsize=(8,5))
sns.barplot(x='Profit',y="Sub-Category",data=product_profit,hue="Profit")
plt.show()
```



Walmart Sales Dashboard

Region

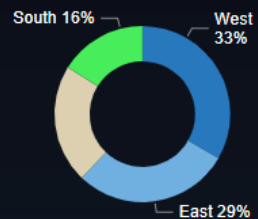
Central

East

South

West

Sales By Region



Sales

1.6M

Profit

175K

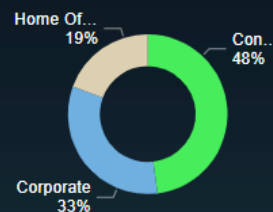
Order

22K

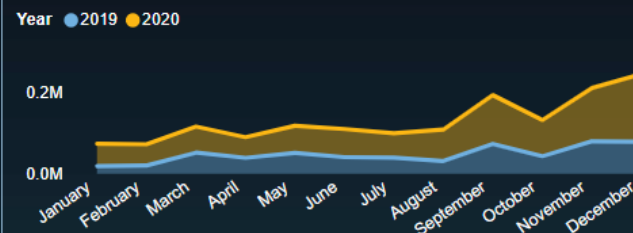
Avg.DeliveryTime
(Days)

4

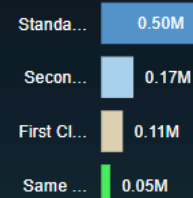
Sales by Customer Segment



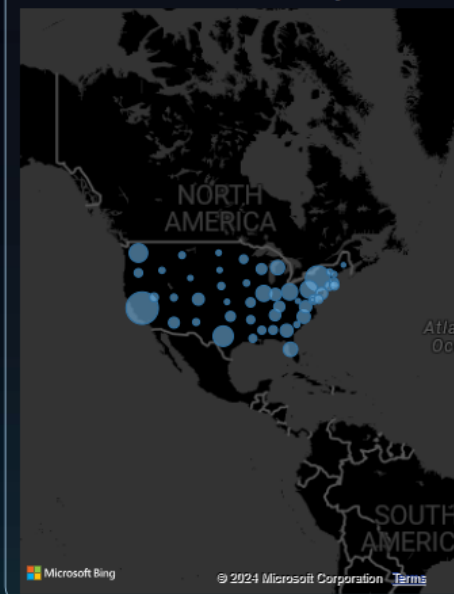
Monthly Sales By Years



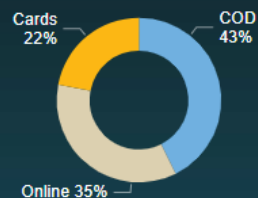
Sales by Ship Mode



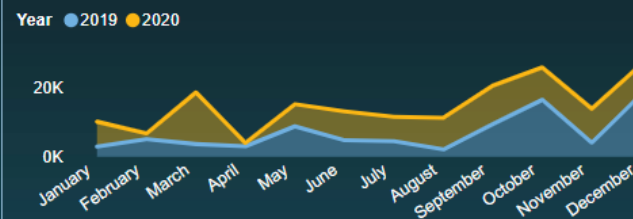
Total Sales and Profit by State



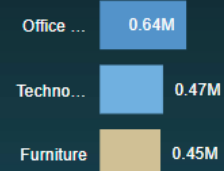
Payment Modes



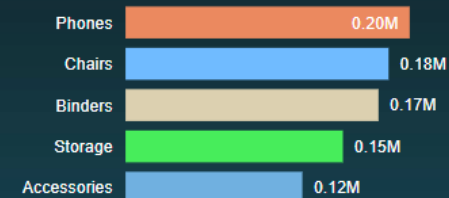
Monthly Profit By Years



Sales by Category



Sales by Sub-Category



Filters

GLOBAL SALES DASHBOARD

Quarter

Select all

Qtr 1

Qtr 2

Qtr 3

Qtr 4

Country

All

Year

All

Month

All

Orders

178K

Sales

13M

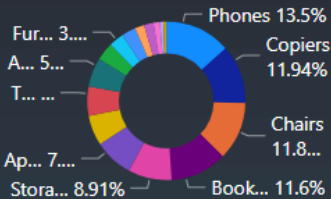
Profit

1M

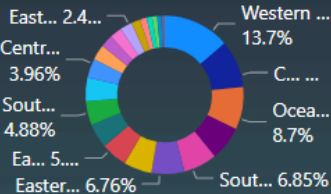
Ship Days

4

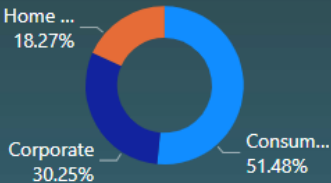
Sales by Sub-Category



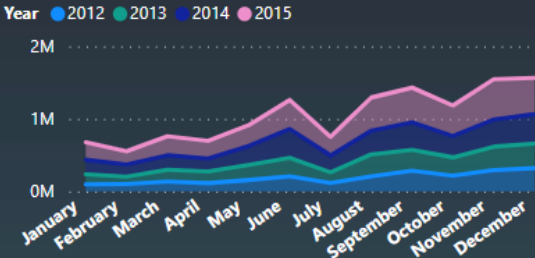
Sales by Region



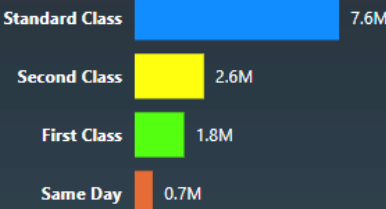
Sales by Segment



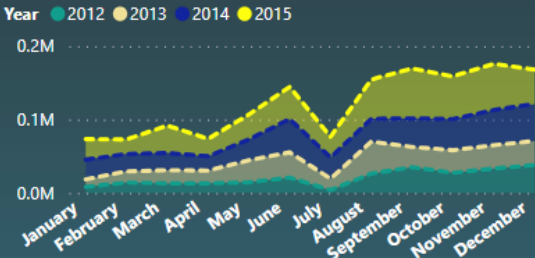
Sales by Month and Year



Sales by Ship Mode



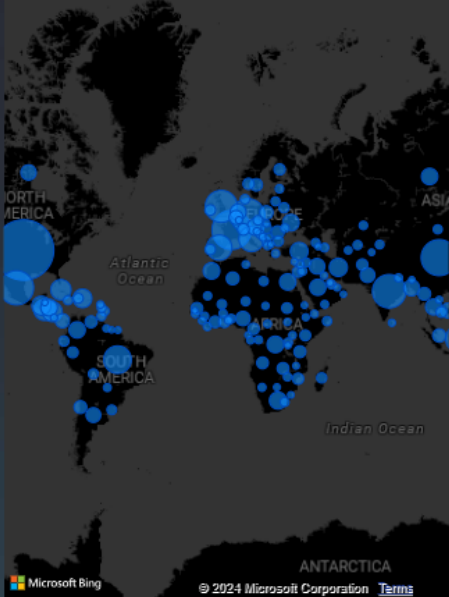
Profit by Month and Year



Sales by Category



Sales by Country



Sales by Market



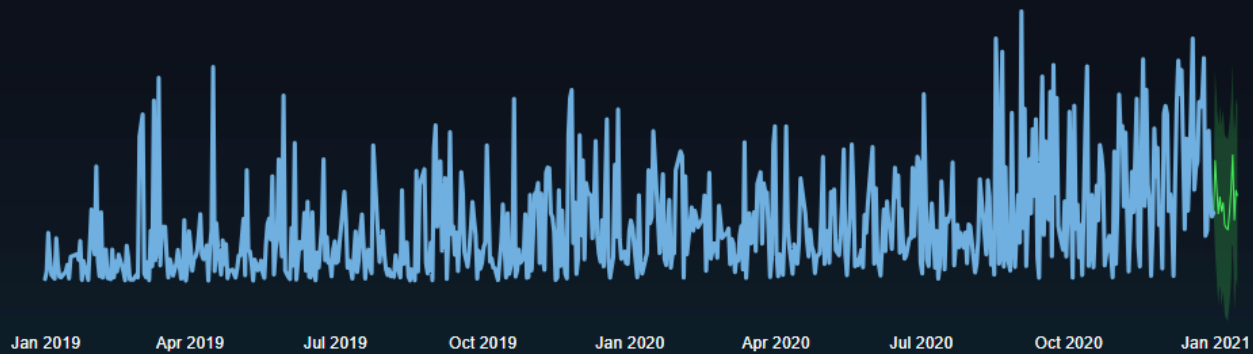
Time-Series Analysis For Sales Forecasting

- ❖ Time series analysis is a statistical technique used to analyze temporal data and make predictions.
- ❖ By analyzing historical sales data, businesses can forecast future sales trends and demand patterns.
- ❖ Time series models such as ARIMA (Auto Regressive Integrated Moving Average) and Exponential Smoothing are commonly used for sales forecasting.



Walmart Sales Forecast - 15 Days

Sales Forecasting - 15 Days



Sales Forecasting - 15 Days



Sales By States



Country

Australia

City

Sydney

Category

Technology

Sub-Category

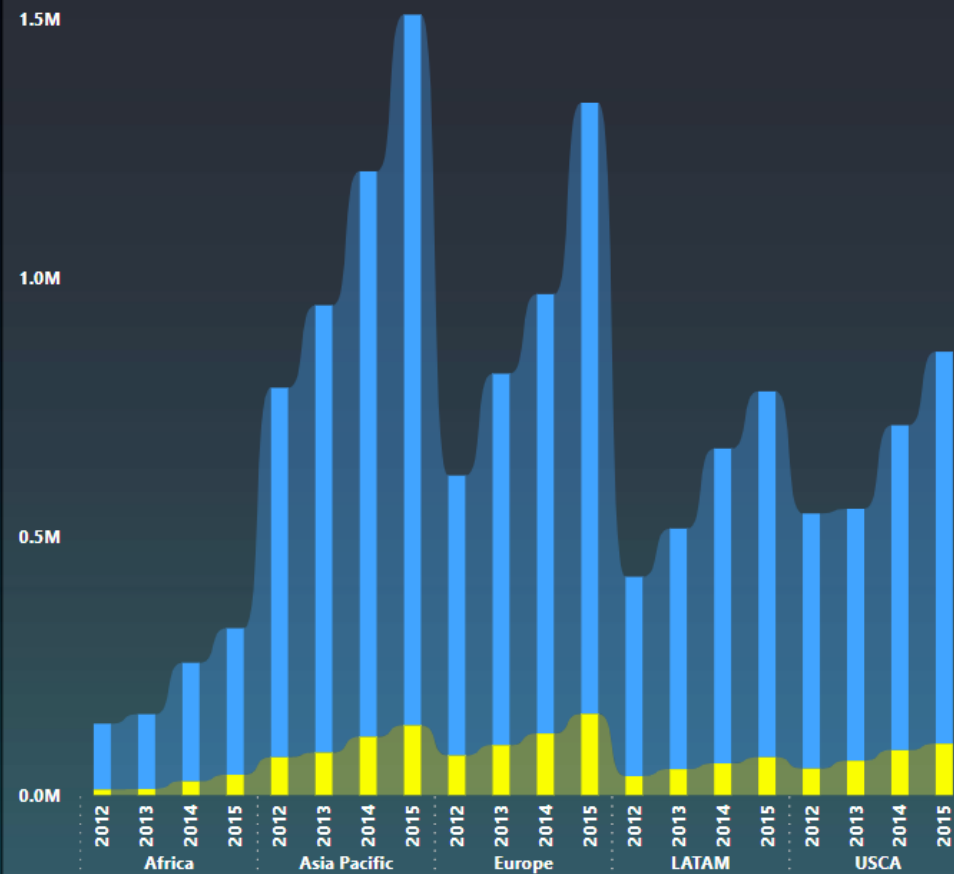


Profit by Country



Sales and Profit Comparison By Year

● Sum of Sales ● Sum of Profit



Actionable Insights and Decision Making

- ❖ The insights generated from data analysis and time series forecasting empower businesses to make informed decisions.
- ❖ By understanding sales trends and forecasting future sales, businesses can optimize pricing strategies, allocate resources efficiently, and identify growth opportunities.



Conclusion

- ❖ Data analysis, Power BI and time series analysis are powerful tools for uncovering sales insights and forecasting future sales.
- ❖ By harnessing the power of data-driven insights, businesses can gain a competitive edge, drive growth, and enhance overall performance.
- ❖ Continued investment in data analytics capabilities and technology is essential for staying ahead in today's dynamic business environment.



Breast Cancer Prediction Using Data Science

- ❖ Welcome to the presentation on Breast Cancer Prediction Using Data Science.
- ❖ Breast cancer is a prevalent disease affecting women worldwide, making early detection crucial.
- ❖ This presentation explores the application of data science techniques for predicting breast cancer.



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Format as Table

Cell Styles

Insert

Delete

Format

Cells

AutoSum

Fill

Clear

Editing

Sort & Filter

Find & Select

Add-ins

Add-ins

A1

id

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	id	diagnosis	radius_me	texture_m	perimeter_	area_mea	smoothne	compactn	concavity	concave p	symmetry_	fractal_dir	radius_se	texture_se	perimeter_	area_se	smoothne	compactn	concavity	concave p	symmetry_	fractal_dir	radius_wo te
2	842302	M	17.99	10.38	122.8	1001	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871	1.095	0.9053	8.589	153.4	0.006399	0.04904	0.05373	0.01587	0.03003	0.006193	25.38
3	842517	M	20.57	17.77	132.9	1326	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	0.5435	0.7339	3.398	74.08	0.005225	0.01308	0.0186	0.0134	0.01389	0.003532	24.99
4	84300903	M	19.69	21.25	130	1203	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999	0.7456	0.7869	4.585	94.03	0.00615	0.04006	0.03832	0.02058	0.0225	0.004571	23.57
5	84348301	M	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744	0.4956	1.156	3.445	27.23	0.00911	0.07458	0.05661	0.01867	0.05963	0.009208	14.91
6	84358402	M	20.29	14.34	135.1	1297	0.1003	0.1328	0.198	0.1043	0.1809	0.05883	0.7572	0.7813	5.438	94.44	0.01149	0.02461	0.05688	0.01885	0.01756	0.005115	22.54
7	843786	M	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613	0.3345	0.8902	2.217	27.19	0.00751	0.03345	0.03672	0.01137	0.02165	0.005082	15.47
8	844359	M	18.25	19.98	119.6	1040	0.09463	0.109	0.1127	0.074	0.1794	0.05742	0.4467	0.7732	3.18	53.91	0.004314	0.01382	0.02254	0.01039	0.01369	0.002179	22.88
9	84458202	M	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451	0.5835	1.377	3.856	50.96	0.008805	0.03029	0.02488	0.01448	0.01486	0.005412	17.06
10	844981	M	13	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389	0.3063	1.002	2.406	24.32	0.005731	0.03502	0.03553	0.01226	0.02143	0.003749	15.49
11	84501001	M	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243	0.2976	1.599	2.039	23.94	0.007149	0.07217	0.07743	0.01432	0.01789	0.01008	15.09
12	845636	M	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697	0.3795	1.187	2.466	40.51	0.004029	0.009269	0.01101	0.007591	0.0146	0.003042	19.19
13	84610002	M	15.78	17.89	103.6	781	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082	0.5058	0.9849	3.564	54.16	0.005771	0.04061	0.02791	0.01282	0.02008	0.004144	20.42
14	846226	M	19.17	24.8	132.4	1123	0.0974	0.2458	0.2065	0.1118	0.2397	0.078	0.9555	3.568	11.07	116.2	0.003139	0.08297	0.0889	0.0409	0.04484	0.01284	20.96
15	846381	M	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338	0.4033	1.078	2.903	36.58	0.009769	0.03126	0.05051	0.01992	0.02981	0.003002	16.84
16	84667401	M	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682	0.2121	1.169	2.061	19.21	0.006429	0.05936	0.05501	0.01628	0.01961	0.008093	15.03
17	84799002	M	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077	0.37	1.033	2.879	32.55	0.005607	0.0424	0.04741	0.0109	0.01857	0.005466	17.46
18	848406	M	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922	0.4727	1.24	3.195	45.4	0.005718	0.01162	0.01998	0.01109	0.0141	0.002085	19.07
19	84862001	M	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356	0.5692	1.073	3.854	54.18	0.007026	0.02501	0.03188	0.01297	0.01689	0.004142	20.96
20	849014	M	19.81	22.15	130	1260	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395	0.7582	1.017	5.865	112.4	0.006494	0.01893	0.03391	0.01521	0.01356	0.001997	27.32
21	8510426	B	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766	0.2699	0.7886	2.058	23.56	0.008462	0.0146	0.02387	0.01315	0.0198	0.0023	15.11
22	8510653	B	13.08	15.71	85.63	520	0.1075	0.127	0.04568	0.0311	0.1967	0.06811	0.1852	0.7477	1.383	14.67	0.004097	0.01898	0.01698	0.00649	0.01678	0.002425	14.5
23	8510824	B	9.504	12.44	60.34	273.9	0.1024	0.06492	0.02956	0.02076	0.1815	0.06905	0.2773	0.9768	1.909	15.7	0.009606	0.01432	0.01985	0.01421	0.02027	0.002968	10.23
24	8511133	M	15.34	14.26	102.5	704.4	0.1073	0.2135	0.2077	0.09756	0.2521	0.07032	0.4388	0.7096	3.384	44.91	0.006789	0.05328	0.06446	0.02252	0.03672	0.004394	18.07
25	851509	M	21.16	23.04	137.2	1404	0.09428	0.1022	0.1097	0.08632	0.1769	0.05278	0.6917	1.127	4.303	93.99	0.004728	0.01259	0.01715	0.01038	0.01083	0.001987	29.17
26	852552	M	16.65	21.38	110	904.6	0.1121	0.1457	0.1525	0.0917	0.1995	0.0633	0.8068	0.9017	5.455	102.6	0.006048	0.01882	0.02741	0.0113	0.01468	0.002801	26.46
27	852631	M	17.14	16.4	116	912.7	0.1186	0.2276	0.2229	0.1401	0.304	0.07413	1.046	0.976	7.276	111.4	0.008029	0.03799	0.03732	0.02397	0.02308	0.007444	22.25
28	852763	M	14.59	21.53	87.41	644.8	0.1054	0.1868	0.1425	0.08783	0.2353	0.06034	0.2545	0.8833	2.11	31.85	0.004453	0.02055	0.02581	0.01353	0.01454	0.002711	17.63

<

>

data

+

100%

Ready

Accessibility: Unavailable

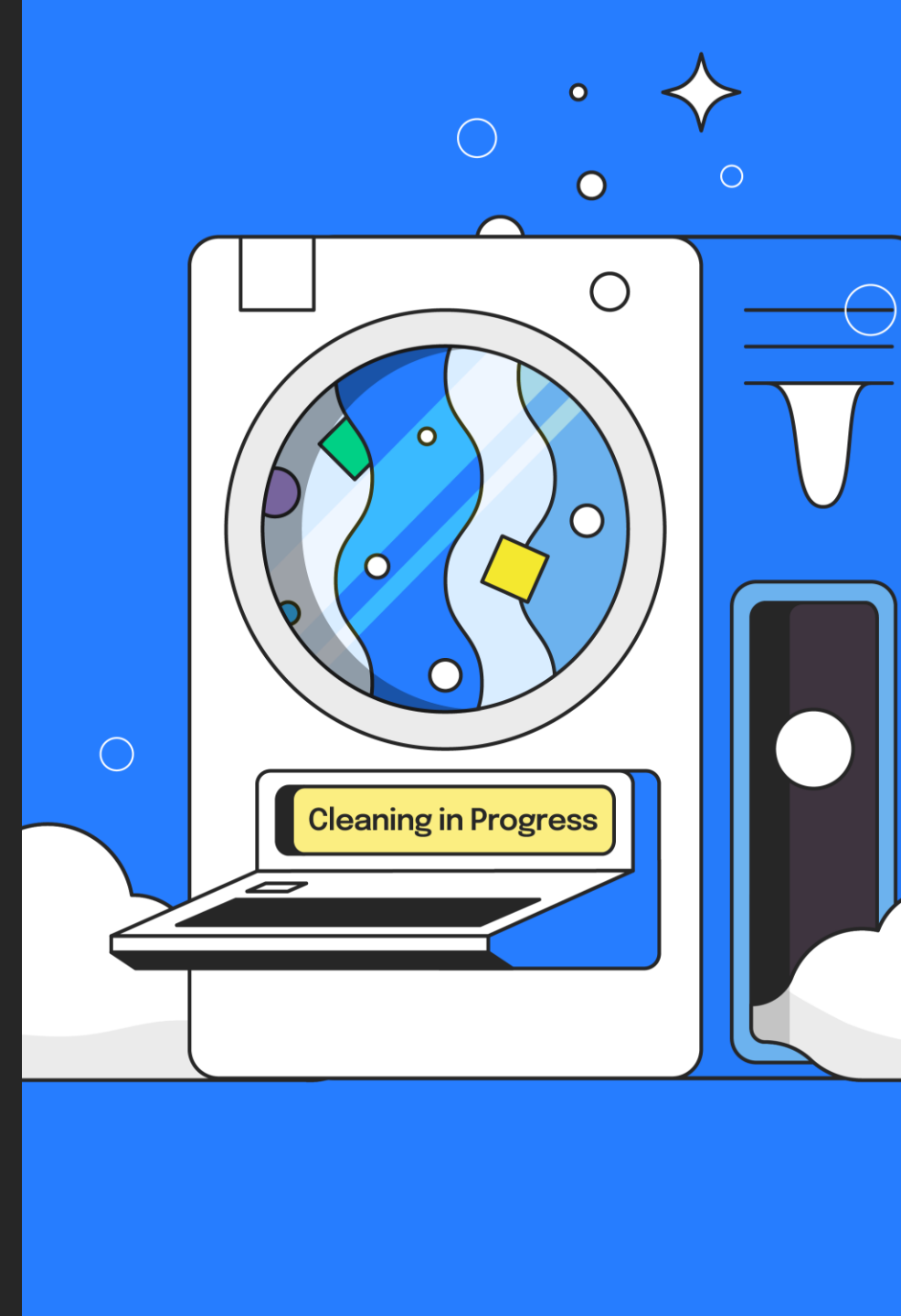
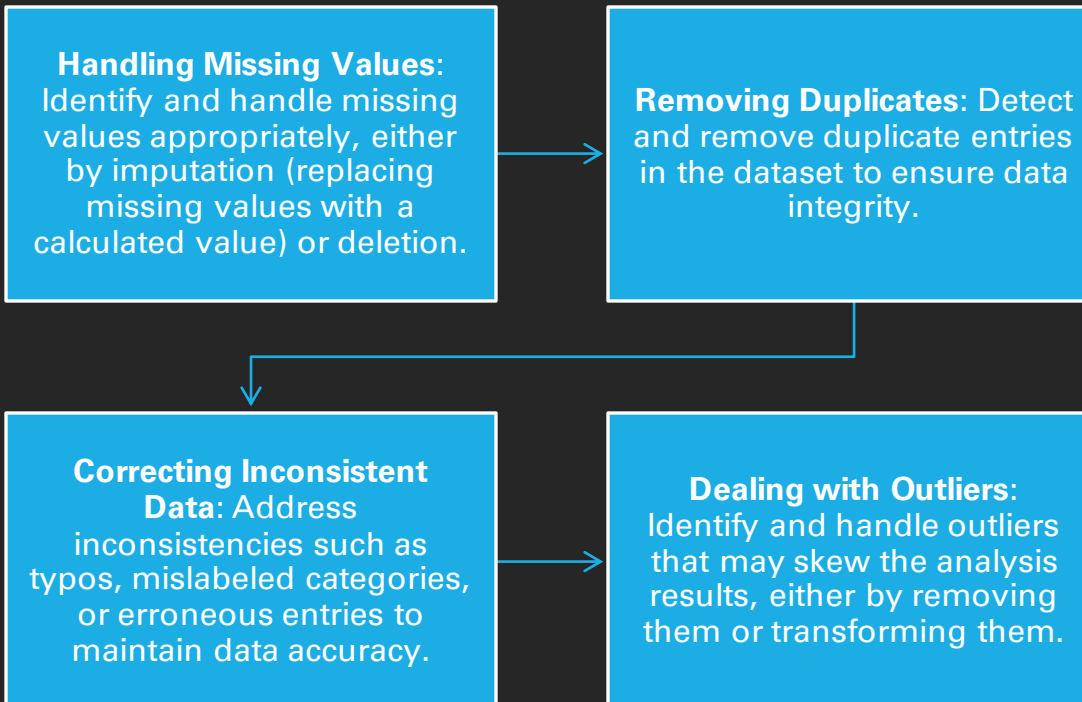
28°C Smoke

Search

ENG IN

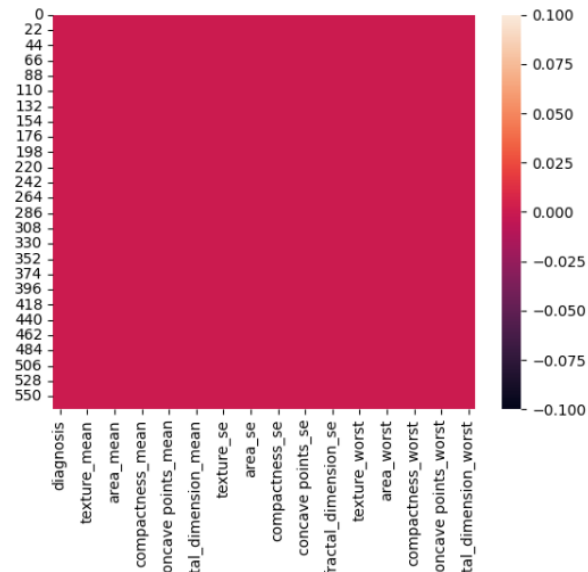
10:18 20-03-2024

Data Pre-processing & Data Cleaning



```
In [14]: sns.heatmap(data.isnull())
```

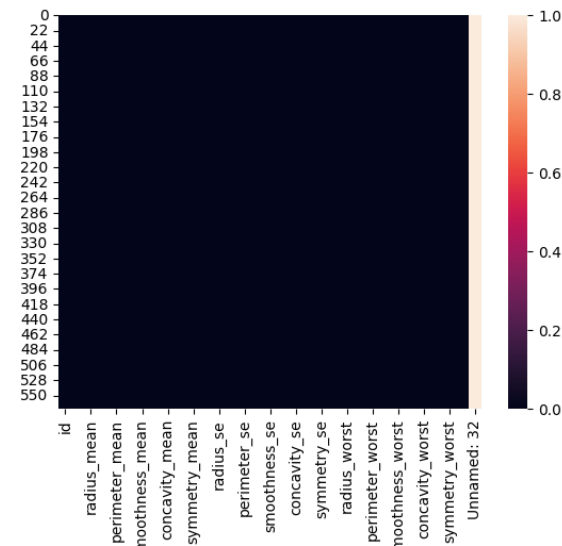
```
Out[14]: <Axes: >
```



Data Pre-processing and Data Cleaning

```
In [9]: sns.heatmap(data.isnull())
```

```
Out[9]: <Axes: >
```



```
1 import pandas as pd
2 from sklearn.preprocessing import StandardScaler
3 from sklearn.model_selection import train_test_split
4 from sklearn.linear_model import LogisticRegression
5 from sklearn.metrics import accuracy_score, classification_report
6 import pickle
7
8
9 def get_clean_data():
10     data = pd.read_csv('data.csv')
11     # clean the data
12     data = data.drop(['Unnamed: 32', 'id'], axis=1)
13     data['diagnosis'] = data['diagnosis'].map({'M':1, 'B':0})
14     print(data.info())
15     return data
16
```

```
fractal_dimension_mean      0
radius_se                   0
texture_se                   0
perimeter_se                 0
area_se                      0
smoothness_se                0
compactness_se               0
concavity_se                 0
concave points_se            0
symmetry_se                  0
fractal_dimension_se         0
radius_worst                 0
texture_worst                 0
```



Model Training and Evaluation

- Machine learning models, such as logistic regression or linear regression, are trained on the preprocessed data.
- Model performance is evaluated using metrics like accuracy, precision.
- Streamlit can display these metrics in real-time, allowing users to understand the model's performance.

Evaluation of the model

```
In [43]: from sklearn.metrics import accuracy_score
```

```
In [45]: Accuracy = accuracy_score(y_test,y_predict)
```

```
In [46]: Accuracy
```

```
Out[46]: 0.9824561403508771
```

```
In [47]: #Classification report
from sklearn.metrics import classification_report
```

```
In [49]: print(classification_report(y_test,y_predict))
```

	precision	recall	f1-score	support
0	0.97	1.00	0.99	108
1	1.00	0.95	0.98	63
accuracy			0.98	171
macro avg	0.99	0.98	0.98	171
weighted avg	0.98	0.98	0.98	171

```
16
17 def create_model(data):
18     x=data.drop('diagnosis',axis=1) # Independent variable
19     y=data['diagnosis'] #dependent variable
20
21     # scale the data
22     scaler = StandardScaler()
23     x=scaler.fit_transform(x)
24
25     # split the data
26     x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.4)
27
28     # train the data
29     model = LogisticRegression()
30     model.fit(x_train,y_train)
31
32     # test the model
33     y_predict = model.predict(x_test)
34     print('Accuracy of model :',accuracy_score(y_test,y_predict))
35     print('Classification report :',classification_report(y_test,y_predict))
36
37     return model,scaler
38
```

Normalize the data

```
In [27]: # Scikit-Learn
from sklearn.preprocessing import StandardScaler

# create a scaler object
scaler = StandardScaler()

# fit the scaler to the data and transform the data
x_scaled = scaler.fit_transform(x)
```

```
In [28]: x_scaled
```

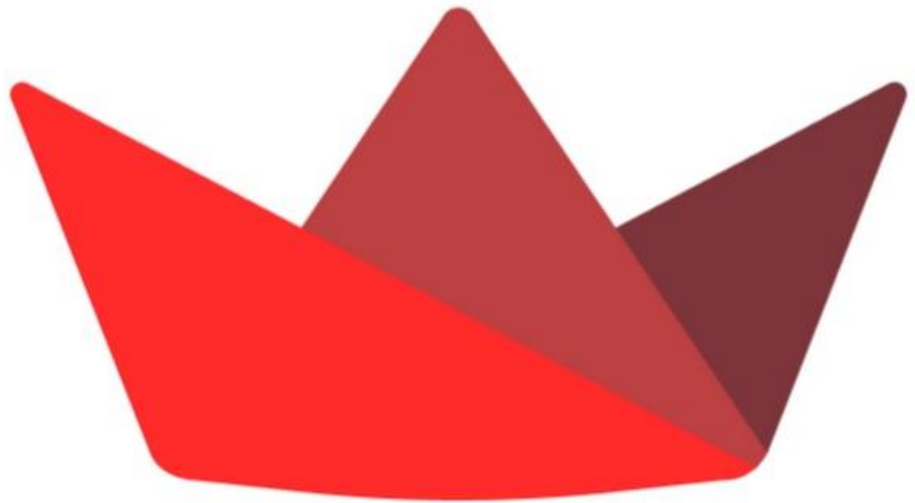
```
Out[28]: array([[ 1.09706398, -2.07333501,  1.26993369, ...,  2.29607613,
  2.75062224,  1.93701461],
 [ 1.82982061, -0.35363241,  1.68595471, ...,  1.0870843 ,
 -0.24388967,  0.28118999],
 [ 1.57988811,  0.45618695,  1.56650313, ...,  1.95500035,
  1.152255 ,  0.20139121],
 ...,
 [ 0.70228425,  2.0455738 ,  0.67267578, ...,  0.41406869,
 -1.10454895, -0.31840916],
 [ 1.83834103,  2.33645719,  1.98252415, ...,  2.28998549,
  1.91908301,  2.21963528],
 [-1.80840125,  1.22179204, -1.81438851, ..., -1.74506282,
 -0.04813821, -0.75120669]])
```

```
None
Accuracy of model : 0.9736842105263158
Classification report :

```

	precision	recall	f1-score	support
0	0.97	0.99	0.98	148
1	0.97	0.95	0.96	80
accuracy		0.97		228
macro avg	0.97	0.97	0.97	228
weighted avg	0.97	0.97	0.97	228

○ PS E:\Internship\Breast Cancer Predictor>



Building The Streamlit Application

- Streamlit applications are built using simple Python scripts.
- Developers can add widgets like sliders, dropdowns, and buttons to create interactive elements.
- Streamlit's reactive framework automatically updates the interface based on user input or changes in the underlying data.

```
Go Run ... ← → Breast Cancer Predictor

main.py app.py × data.csv style.css

app.py > main
1 import streamlit as st
2 import pickle
3 import pandas as pd
4 import plotly.graph_objects as go
5 import numpy as np
6
7 def get_clean_data():
8     data = pd.read_csv('data.csv')
9     # clean the data
10    data=data.drop(['Unnamed: 32','id'],axis=1)
11    data['diagnosis'] = data['diagnosis'].map({'M':1,'B':0})
12    print(data.info())
13    return data
14
```

```
def add_predictions(input_data):
    model = pickle.load(open('E:\Internship\Breast Cancer Predictor\model.pkl','rb'))
    scaler = pickle.load(open("E:\Internship\Breast Cancer Predictor\scaler.pkl","rb"))

    # we convert input_data dictionary in array (in two dimensional)
    input_array = np.array(list(input_data.values())).reshape(1,-1)

    # Scale the input_array using scaler
    input_array_scaled = scaler.transform(input_array)

    st.subheader("Cell Cluster Prediction")
    st.write("The cell cluster is:")

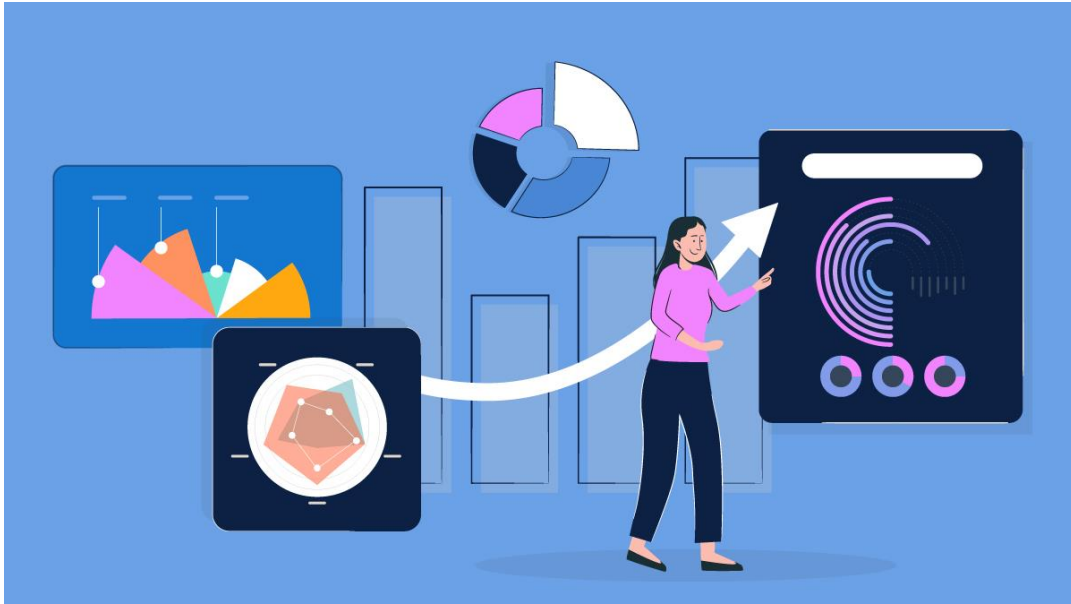
    # now we make prediction using input_array_scaled
    prediction = model.predict(input_array_scaled)
    if prediction[0]==0:
        st.write("<span class='diagnosis benign'>Benign</span>", unsafe_allow_html=True)
    else:
        st.write("<span class='diagnosis malignant'>Malignant</span>", unsafe_allow_html=True)

    # probability values for prediction(benign or malignant)
    st.write('Probability of being Benign:',model.predict_proba(input_array_scaled)[0][0])
    st.write('Probability of being Malignant:',model.predict_proba(input_array_scaled)[0][1])

    st.write('This app can assist medical professionals in making a diagnosis, but should not be used as a substitute for a professional
```

```
18
19 data = get_clean_data()
20
21 slider_labels = [
22     ("Radius (mean)", "radius_mean"),
23     ("Texture (mean)", "texture_mean"),
24     ("Perimeter (mean)", "perimeter_mean"),
25     ("Area (mean)", "area_mean"),
26     ("Smoothness (mean)", "smoothness_mean"),
27     ("Compactness (mean)", "compactness_mean"),
28     ("Concavity (mean)", "concavity_mean"),
29     ("Concave points (mean)", "concave points_mean"),
30     ("Symmetry (mean)", "symmetry_mean"),
31     ("Fractal dimension (mean)", "fractal_dimension_mean"),
32     ("Radius (se)", "radius_se"),
33     ("Texture (se)", "texture_se"),
34     ("Perimeter (se)", "perimeter_se"),
35     ("Area (se)", "area_se"),
36     ("Smoothness (se)", "smoothness_se"),
37     ("Compactness (se)", "compactness_se"),
38     ("Concavity (se)", "concavity_se"),
39     ("Concave points (se)", "concave points_se"),
40     ("Symmetry (se)", "symmetry_se"),
41     ("Fractal dimension (se)", "fractal_dimension_se"),
42     ("Radius (worst)", "radius_worst"),
43     ("Texture (worst)", "texture_worst"),
44     ("Perimeter (worst)", "perimeter_worst"),
45     ("Area (worst)", "area_worst"),
46     ("Smoothness (worst)", "smoothness_worst"),
47     ("Compactness (worst)", "compactness_worst"),
48     ("Concavity (worst)", "concavity_worst"),
49     ("Concave points (worst)", "concave points_worst"),
50     ("Symmetry (worst)", "symmetry_worst"),
51     ("Fractal dimension (worst)", "fractal_dimension_worst"),
52 ]
53
54 input_dict = {}
55
56 for label,key in slider_labels:
57     input_dict[key]=st.sidebar.slider(
```

Visualizing Predictions



- Streamlit enables developers to visualize predictions using plots, charts, and tables.
- Interactive visualizations help users understand the factors influencing the predicted outcome.
- Streamlit's built-in support for popular plotting libraries like Matplotlib and Plotly makes it easy to create engaging visualizations.

```

64
65 def get_scaled_values(input_dict):
66     data = get_clean_data()
67
68     X = data.drop(['diagnosis'], axis=1)
69
70     scaled_dict = {}
71
72     for key, value in input_dict.items():
73         max_val = X[key].max()
74         min_val = X[key].min()
75         scaled_value = (value - min_val) / (max_val - min_val)
76         scaled_dict[key] = scaled_value
77
78     return scaled_dict
79
80

```

```

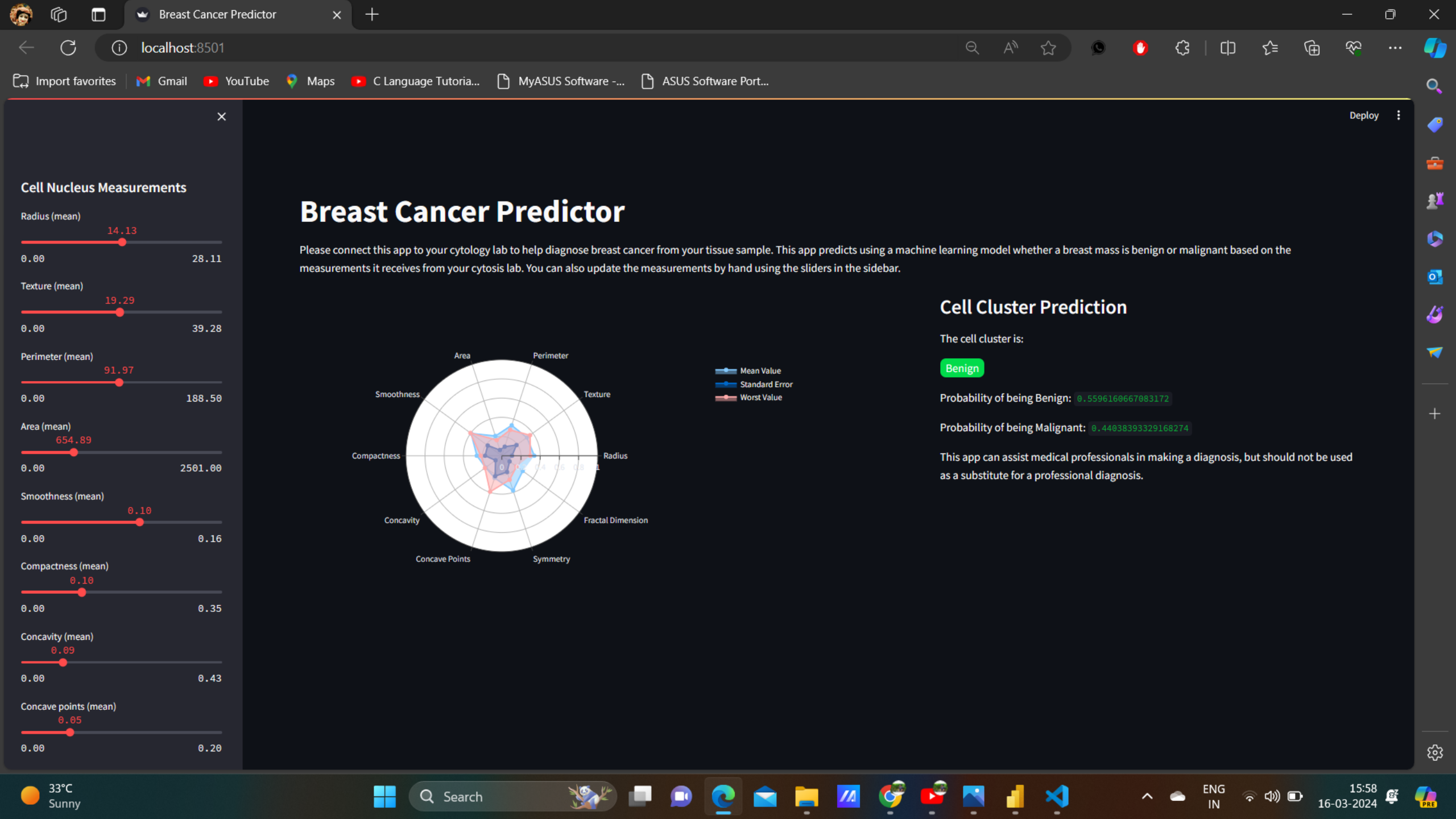
163
164 def main():
165     st.set_page_config(
166         page_title='Breast Cancer Predictor',
167         page_icon=':Female Doctor:',
168         layout='wide',
169         initial_sidebar_state='expanded'
170     )
171
172     with open("E:\Internship\Breast Cancer Predictor\style.css") as f:
173         st.markdown("<style>{}</style>".format(f.read()), unsafe_allow_html=True)
174     input_data = add_sidebar()
175
176     #st.write(input_data)
177
178     with st.container():
179         st.title("Breast Cancer Predictor")
180         st.write("Please connect this app to your cytology lab to help diagnose breast cancer from your tissue sample. This app predicts using a machine learning model.")
181
182     col1,col2 = st.columns([15,10])
183
184     with col1:
185         radar_chart = get_radar_chart(input_data)
186         st.plotly_chart(radar_chart)
187     with col2:
188         add_predictions(input_data)
189
190
191 if __name__ == '__main__':
192     main()

```

```

81
82 def get_radar_chart(input_data):
83
84     input_data = get_scaled_values(input_data)
85
86     categories = ['Radius', 'Texture', 'Perimeter', 'Area',
87                  'Smoothness', 'Compactness',
88                  'Concavity', 'Concave Points',
89                  'Symmetry', 'Fractal Dimension']
89
90
91     fig = go.Figure()
92
93     fig.add_trace(go.Scatterpolar(
94         r=[
95             input_data['radius_mean'], input_data['texture_mean'], input_data['perimeter_mean'],
96             input_data['area_mean'], input_data['smoothness_mean'], input_data['compactness_mean'],
97             input_data['concavity_mean'], input_data['concave points_mean'], input_data['symmetry_mean'],
98             input_data['fractal_dimension_mean']
99         ],
100         theta=categories,
101         fill='toself',
102         name='Mean Value'
103     ))
104
105     fig.add_trace(go.Scatterpolar(
106         r=[
107             input_data['radius_se'], input_data['texture_se'], input_data['perimeter_se'], input_data['area_se'],
108             input_data['smoothness_se'], input_data['compactness_se'], input_data['concavity_se'],
109             input_data['concave points_se'], input_data['symmetry_se'], input_data['fractal_dimension_se']
110         ],
111         theta=categories,
112         fill='toself',
113         name='Standard Error'
114     ))
115
116     fig.add_trace(go.Scatterpolar(
117         r=[
118             input_data['radius_worst'], input_data['texture_worst'], input_data['perimeter_worst'],
119             input_data['area_worst'], input_data['smoothness_worst'], input_data['compactness_worst'],
120             input_data['concavity_worst'], input_data['concave points_worst'], input_data['symmetry_worst'],
121             input_data['fractal_dimension_worst']
122         ],
123         theta=categories,
124         fill='toself',
125         name='Worst Value'
126     ))
127
128     fig.update_layout(
129         polar=dict(
130             radialaxis=dict(
131                 visible=True,
132                 range=[0, 1]
133             )),
134         showlegend=True
135     )

```

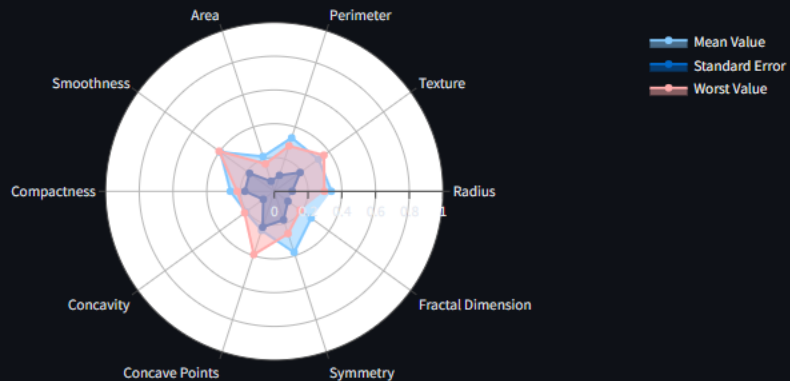


Cell Nucleus Measurements



Breast Cancer Predictor

Please connect this app to your cytology lab to help diagnose breast cancer from your tissue sample. This app predicts using a machine learning model whether a breast mass is benign or malignant based on the measurements it receives from your cytolysis lab. You can also update the measurements by hand using the sliders in the sidebar.



Cell Cluster Prediction

The cell cluster is:

Benign

Probability of being Benign: 0.5596160667083172

Probability of being Malignant: 0.44038393329168274

This app can assist medical professionals in making a diagnosis, but should not be used as a substitute for a professional diagnosis.

Conclusion

- Streamlit provides a powerful platform for developing interactive web applications for breast cancer prediction.
- By combining data science techniques with Streamlit's intuitive interface, developers can create user-friendly tools for healthcare professionals and patients.
- Continued research and innovation in this field hold the promise of improving early detection and treatment of breast cancer.



Tools & Technology Used In Projects

- Visual Studio (Vs Code)
- Python (with Different libraries)
- Machine Learning
- Streamlit
- Power BI
- Jupyter Notebook
- Kaggle (for datasets)





**THANK
YOU**



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